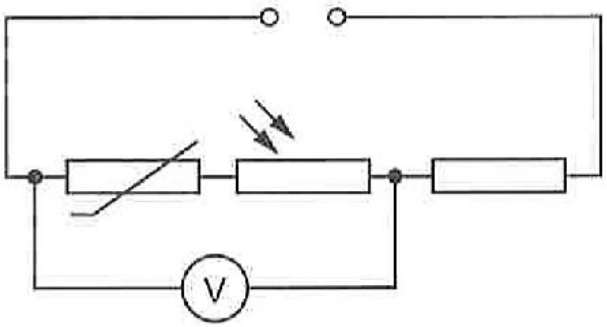
22 A negative temperature coefficient (NTC) thermistor, a light-dependent resistor (LDR) and a fixed resistor are connected in series to a power supply. A voltmeter is placed across the NTC-LDR combination.



What conditions of brightness and temperature will produce the largest reading on the voltmeter?

	temperature	brightness
A	high	high
B	high	low
C	low	high
D	low	low

L2 **Answer: D**

To get largest p.d. across the voltmeter,
the p.d. across the fixed resistor must be the smallest.

Therefore, apply Potential Divider Principle,
the combined resistance of the thermistor and LDR must be the largest.

The conditions that will result in this are:
the resistance of the thermistor must be the high, so the brightness must be low, and
the resistance of the LDR must be low, so the temperature must be low.